



Patent Filing: CLDI - Collapse Language Divergence Index

Title:

CLDI: Collapse–Language Divergence Index

A Universal Diagnostic for Symbolic System Decay

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♦ Abstract

A diagnostic framework for detecting symbolic system decay across recursive and ritualized systems, including ancient inscriptions, AI-generated outputs, bureaucratic protocols, and institutional documentation. The CLDI (Collapse–Language Divergence Index) scores symbolic content across five core dimensions: **Legibility Degradation**, **Redundancy Frequency**, **Toolcare Deterioration**, **Glyph Simplification**, and **Contextual Anchoring**. The CLDI can be applied to assess semantic drift, loss of volitional integrity, or collapse phenomena in systems where form persists in the absence of functional intent.

♦ Claims

1. A universal scoring framework for diagnosing symbolic drift, comprising a five-dimensional rating system evaluating:

- (a) Legibility degradation
- (b) Redundancy frequency
- (c) Toolcare deterioration
- (d) Glyph simplification
- (e) Contextual anchoring

2. The method of Claim 1, wherein the symbolic material is derived from archaeological inscriptions, including but not limited to undeciphered scripts (e.g., Linear A).

- 3. The method of Claim 1, wherein the symbolic material is generated by recursive artificial intelligence models (e.g., large language models), allowing for detection of semantic drift or model collapse.
- 4. The method of Claim 1, wherein redundancy frequency is calculated using entropy analysis, sequence overlap density, or repeat-pattern detection.
- 5. A software implementation of the method of Claim 1, employing natural language processing (NLP) and/or computer vision (CV) modules to automatically score symbolic content based on the five dimensions.
- 6. A diagnostic dashboard that visualizes CLDI scores as heatmaps, radar charts, and decay trajectories, enabling longitudinal monitoring of symbolic system degradation.
- 7. A pattern-matching architecture capable of detecting symbolic mimicry, defined as the presence of consistent structural form in the absence of volitional or semantic intent.
- 8. A semiotic forensics toolkit incorporating the CLDI system for use across diverse domains including archaeological decipherment, AI alignment, and bureaucratic process audits.

◆ **Description**

The CLDI is a recursive-aware, semiotic decay diagnostic developed for assessing symbolic systems under collapse pressure. It originated in archaeological research analyzing Linear A tablets, where bureaucratic residues preserved in clay were scored using a 5-dimensional decay rubric. The same rubric proved transferable to AI model outputs and institutional documents.

Core Dimensions:

Dimension	Meaning
Legibility Degradation	Physical erosion, visual degradation, or stylus neglect
Redundancy Frequency	Repetition of patterns or sequences indicating collapse-phase drift
Toolcare Deterioration	Instrument fatigue, apathy signals, lack of precision
Glyph Simplification	Reduction in symbolic complexity over time
Contextual Anchoring	Fixed symbolic roles losing relevance or functional grounding

The scoring framework enables both human and automated analysis. For example:

- **AI Drift:** Recursive LLMs producing tautological or degraded outputs.
- **Archaeological Inscription:** Linear A tablets showing ritualized formats, tool apathy, and rigid position clustering.
- **Bureaucracy:** Institutions generating forms and checklists with decoupled semantics.

Implementations:

- Python toolkits using OpenCV (for visual degradation) and NLTK/transformers (for text redundancy).
- CLDI dashboards aggregating scores across documents or time-series data.
- White paper outputs for timestamped analyses of symbolic entropy and collapse behavior.

Figures (not shown here):

- Fig. 1: CLDI scoring rubric
- Fig. 2: Scored HT20 tablet example (overlay)
- Fig. 3: Radar chart of CLDI profiles
- Fig. 4: Cross-domain comparison (AI, bureaucracy, archaeology)